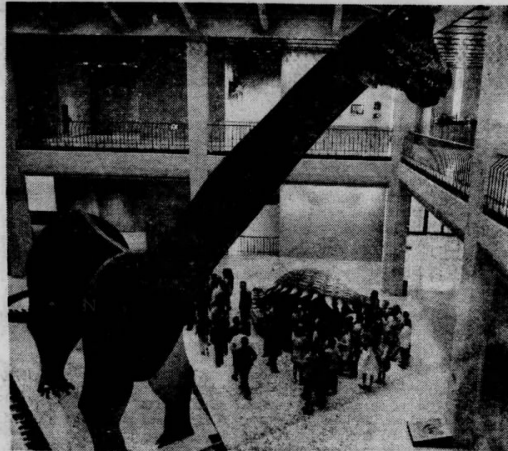


This is the shape
Diplodocus will take
when he's finally
hung together.



Out at the Museum
Science a prehistoric
creature is taking shape
One thing is certain
it is not a bit like
playing with tinker

How to build a dinosaur



By CHARLOTTE PHELAN
Post Reporter

Visitors to the Houston Museum of Natural Science have been aware for several years that a mammoth, complicated project involving some massive, petrified bones is in progress there on the first floor.

After all, the only way to assemble the 75-foot skeleton of a dinosaur — a Diplodocus, to be exact — is to bring the pieces back together on the site of permanent display, because a restored dinosaur skeleton just isn't very portable.

Nor do you put one together very quickly. Ask the museum's director, Dr. Thomas E. Pulley, about that. He has long since grown accustomed to jibes about the time already spent on the dinosaur, even as he projects another year's work—at least—before the prehistoric bones are all in place.

However, when one talks in terms of 150 million years, the estimated age of the Diplodocus, what are a mere five?

As Dr. Wann Langston, Jr., of the University of Texas, the vertebrate paleontologist in charge of the endeavor, has pointed out, there are numerous problems not the least of which is the sheer engineering challenge of putting those bones together in such a way that they will stay together.

Dr. Pulley just grins when the protraction of his project is mentioned, but then he becomes serious.

"This isn't like building a school or a church, where you can hire people to do something they've spent all their lives doing. Anything anybody has been doing on this he's doing for the first and last time. At least, I'm sure we're not going to build any more dinosaurs."

The Diplodocus, Dr. Pulley explained, is the commonest of dinosaurs. When it is finally mounted, the skeleton will be more than 20 feet high and 70 to 75 feet long, although its lengthy tail will be curled like a question mark to keep it from taking up too much space.

Dr. Pulley's account of the acquisition of his prize goes back to "the turn of the century, when Andrew Carnegie had made his fortune and decided he would build a museum.

"He wanted something everybody would take notice of, and nothing was more exciting at that time than big dinosaurs. He sent several large crews of men out West, to Colorado, Utah and Wyoming.

"Those field crews sent box after box of dinosaur material back to Carnegie Institute in Pittsburgh, and they assembled one of the first big complete skeletons of a Diplodocus, one of the long-

necked and osaurus.

"When the had mounted made plaster reproductions them to muse Japan.

"Their field of other Di were just put tee you nob Diplodocus.

"From 19 rested in t Institute. Th traded a for them. Th plans change skeleton, the \$15,000. The that money.

"The bone shape they w from Wyoming still on them ed off. Our M to pay for t aged bones. ones. "Like bones are r complete." I on the slo enormous a with suspens rose high ove massive leg place. "We bone separate major job in the forging bones, and v smith shop o this is being

"The idea strong enoug still be as e that the sh ports—is whi

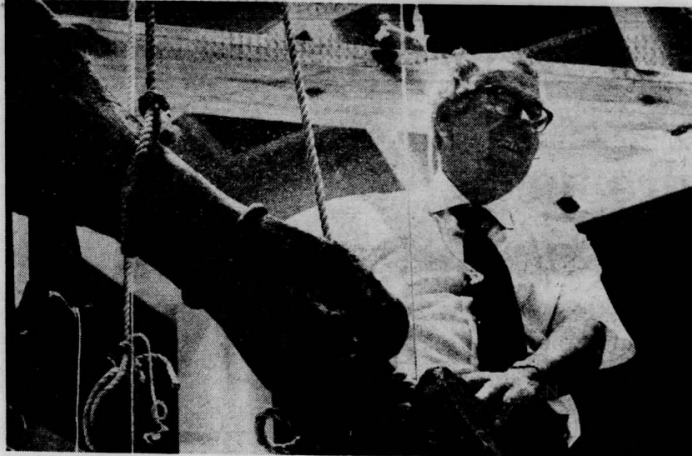
"These bot to stone, you there wouldn for them to i is now a whi ing from min

"Mineral-b organic stuff mineral depo material. Th accurate that will look exa

"We are v Langston in ery bone of t laboratory o personally su exhibit. He l

"We've ha Dale Reimer

of Natural ic hape. : Building toys.



Dr. Pulley
bought
Diplodocus
in crates.

Some of
Diplodocus'
bones are
missing.

long-tailed herbivorous din-
people at Carnegie Institute
their skeleton, they then
molds from which concrete
could be cast. They sent
seums in Europe and even

d men also brought in a lot
plodocus bones, and these
it in storage. I can guaran-
only needs more than one

05 to 1960 these bones
he basement at Carnegie
en the Cleveland Museum
collection of bird skins
en the Cleveland Museum's
d and they did not want the
ey offered it to me for
Junior League here put up

s were still in the same
ere when they were shipped
ng. There was a lot of rock
, which needed to be clean-
useum Guild put up \$10,000
hat and for restoring dam-
and replacing the missing
all old skeletons, some
issing and many are in-
r. Pulley was standing now
ing platform, where an
rangement of scaffolding
sion ropes and hoists
thead and held some of the
bones of the dinosaur in
egan by suspending each
ely with block and tackle. A
assembling the skeleton is
of the iron to support the
ve have a complete black-
ut there at the back where
done.

is to have the supports
n to hold all that weight but
mistrusive as possible, so
keleton—and not its sup-
t you will see.
es have completely turned
understand. If they hadn't,
t have been anything there
ick up. The marrow cavity
te crystalline quartz result-
eral replacement.

earing water flowing over
like bone or a tree leaves a
sit in place of the original
e displacement can be so
a bone can be bored and it
gly like the live bone.
ry fortunate in having Dr.
this part of the world. Ev-
hat animal has been in his
er in Austin where he has
persived its preparation for
as been directing our em-
hout the project.

d people living over there.
was there about two years

so Langston could show him what to do
with each bone. Langston is really
amazing. You can hand him an odd
piece of bone, and he will tell you ex-
actly where it belongs."

Dr. Langston, a modest man, placed
his prowess on a less exalted plane when
he was sought out a few days later in
Austin, where he divides his time be-
tween his laboratory at the University of
Texas' Balcones Research Center and
the Texas Memorial Museum.

"Each one of these projects (dinosaur
skeletons) is something you play by ear.
You can't get a book and read about it.
My own expertise comes from a certain
amount of experience with smaller
things.

"I have seen all the big dinosaurs in
the world, in America and Western Eu-
rope. Of the known procedures for as-
sembling one, each one presents unique
problems.

"I can say this is the only one ever
assembled by computer. We have used a
computer program to determine the re-
quired strength of the metal for support-
ing the bones.

"The engineering department had a
program for determining the strengths
required for a cantilever bridge. We
were able to adapt that to our din-
osaur."

Dr. Langston produced the computer's
"instructions"—page after page of fig-
ures, which needed specialized interpre-
tation. Then he produced a wry chuckle.

"I can say this, if the computer is
wrong, we're in trouble.

"Things would be easier, of course, if
we weren't trying to make the support-
ing framework as inconspicuous as pos-
sible. As it is, what we have immediate-
ly is an engineering problem with no
pre-determined answers.

"Also, you have to remember that
while fossil bones are very heavy, they
are also very fragile and brittle and
have to be handled very carefully.

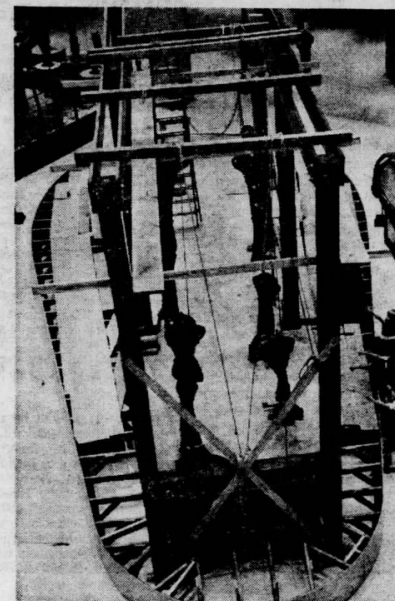
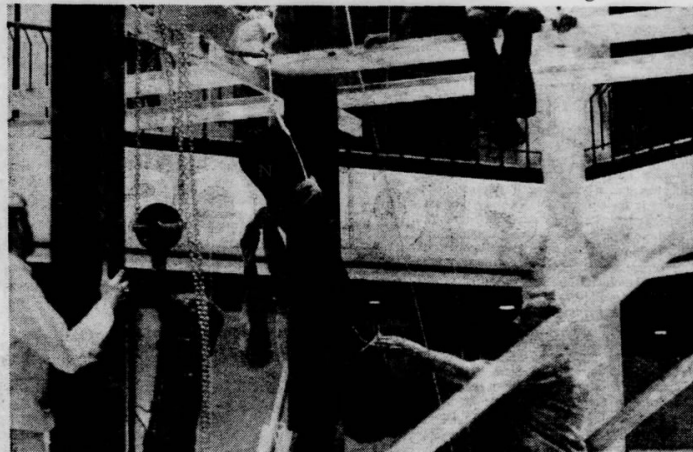
"We have fabricated a steel tube that
conforms fairly closely to the con-
figuration of the bottom bones (of the
spinal column) and made a trough or
shelf on which the bones rest. The bones
will be held in place by metal clamps or
hooks, which we'll have to manufacture.

"Then we will have vertical posts as
supports between the front legs and the
hind legs.

"If we were to pile up all the fossil
bone that goes into the skeleton, it would
weigh a ton and a half or two tons," Dr.
Langston said in answer to a question.

He pointed out, as Dr. Pulley had,
that a number of the bones of the dip-
lodocus were missing, particularly the
rib bones, which are seemingly the least
durable. Also, there are many other

Please see Dinosaur/page 30



Dr. Langston helps
steer leg bone
of Diplodocus into
proper place.

Spotlight — Page 19

1973, THE HOUSTON POST

Broadway

A Little Night Music—Glynis Johns, Len Cariou, Hermione Gingold; elegant new musical about various problems in turn-of-the-century Sweden; Shubert (available).
A Streetcar Named Desire—Rosemary Harris, James Farentino, Vivian Reamon; Repertory Theater of Lincoln Center's revival of Tennessee Williams' drama of disintegration of a Southern belle; (subscription but available).

Cyrano—Christopher Plummer; musical version of Rostand's "Cyrano de Bergerac"; romantic drama about the French soldier-poet with the big nose; Palace (available).
For a Few Dollars More—Micki Grant; all-black musical revue; Edison (available).
Finch—Teaches—Barbara Bel Geddes, Robert Lansing; comedy of cannibalized animals at home of college professor; Plymouth (available).

Grease—Barry Baskwick, Carole Demas; musical extravaganza about youth in the 1950s; Broadway (available).
Irene—Debbie Reynolds; revival of 1919 musical hit about star Irish girl winning a millionaire; large advance.
Jesus Christ Superstar—Jeff Fennell; Rock musical about last days of Christ; Mollinger (available).
Nash—Al Niles—E. G. Marshall; revue by late Ozzie Nash based on his humorous verse; Hayes (available).

Pippin—Ben Vereen, John Rubenstein; musical of medieval youth seeking fulfillment; Imperial (available).
Sweeney—Michael Lee, Ken Howard; musical version of "Two for the Sweeney"; romance of married Midwestern lawyer and an earthy New York girl; Uris (available).
Sleuth—Patrick MacNee; London mystery hit with tricky ending; Music Box (available).
Smith—Don Murray; life of talented takes new turns when his diet mixed up with a musical; Eden (available).
Super—Robert Morris, Tony Roberts, Cyril Richard; musical about two musicians who make a career as girls' basketball team; Booth.

The Championship Season—Richard Dysart, Charles Durning; 1972's best play; from Off-Broadway; bitter review of championship high school sports; Booth.

The Champagne Room—All male cast of 22; earthy slice-of-life drama (same revival) in locker room of British football club before and after a match; Morosco (available).

The Play's The Thing—Hugh Franklin, Elizabeth Owens; revival of Mollat comedy of couple beset by New York City problems and financial ails; Bilox (available).

The Prisoner of Second Avenue—Hector Elizondo, Phillip Newman; Neil Simon comedy about a couple who die with expensive apartment; O'Neil (available).

The River Niger—Douglas Turner Ward; tensions in Negro family headed by father with poetic aspirations; Altkinson (available).

The Sunshine Boys—Jack Albertson, Sam Levene; Neil Simon's newest comedy; two veteran vaudevillians get into a final time; Broadway (available).

The Women—Shirley Fleming, Myrna Loy, Alexis Smith; revival of Clare Booth Luce's 1936 satirical hit about Park Avenue female types; 44th Street (available).

No Broadway openings May 27-June 2, opening May 12—"Cyrano" (Palace), "Sweeney" (Imperial), "Jesus Christ Superstar" (Mollinger), "Finch" (Plymouth), "For a Few Dollars More" (Edison), "Irene" (Hayes), "Nash" (Uris), "Pippin" (Imperial), "Sleuth" (Music Box), "Smith" (Eden), "Super" (Booth), "The Championship Season" (Booth), "The Champagne Room" (Booth), "The Play's The Thing" (Bilox), "The Prisoner of Second Avenue" (O'Neil), "The River Niger" (Altkinson), "The Sunshine Boys" (Broadway), "The Women" (44th Street).

Year of the cop

Continued from page 7.

Doubleday decided against publishing it. It is the story of William Phillips, the Knapp Commission's star witness, who, as a patrolman, was the shining example of a corrupt cop. Phillips also has a murder indictment against him and although his first trial ended in a hung jury the district attorney still intends to prosecute. Doubleday reportedly withdrew its option despite a fat advance to Leonard Shecter, and now Shecter's agent, Theron Raines, says Putnam & Berkley has agreed to publish it.

"I suppose they (Doubleday) have a stuffy image," Raines said, "but we don't have any intention of giving them the money." Shecter has done a brilliant job of describing a crooked cop. Phillips is not a nice guy and the funny thing is when he wanted to be a good cop he solved cases better than anyone. That's why he got to be a detective . . . but if the young cops want to know about graft, it's in this book. It shows how Phillips took his first \$5.

Raines is also the literary agent for Peter Hellman, who is writing the biography of retired Chief of Detectives Albert Seedman.

"I guess you might say I've got the best cop and the worst one," Raines chuckled.

Among the literati there seems to be general agreement that Seedman's experience covering 30 years of detective work has to be fascinating reading. Its title is "The Chief."

"I'm presenting each episode as a mystery and the way Seedman solved the case," says author Hellman. "It's a survey of 10 or 20 great cases which he solved and hundreds of others thrown in. He never talked to anyone about his experiences and in the police department no one is allowed to discuss this sort of thing. A lot of things never came out. He was in charge of the Kitty Genovese case. Thirty-nine people heard her cries for help but no one tried to help her. He handled the investigation of Joe Columbo's attempted assassination and the assassination of Joey Gallo. There's only one view when you talk about cases like that and that's the view from the top."

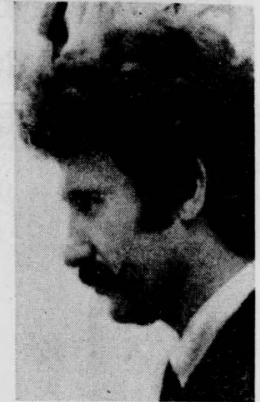
Writers and publishers have been busy going back and forth to libel lawyers since the books by Maas, Whittemore, Daley, Shecter and Hellman are drawn from real life, but that's not the case when it comes to fiction.

Along with Dorothy Uhnak in an unoccupied office of Simon & Schuster, there isn't the slightest hint she spent 14 years "on the job" as a policewoman for the New York City Transit Authority. She is 42, blue-eyed and her hair is sprinkled with gray. Her thin lips curl into a gentle smile when her book "Law and Order" is referred to as Faulkneresque.

"He's one of my heroes, but I don't dare read him or any other great writer when I'm working. They intimidate me."

Dorothy started working on "Law and Order" three years ago, and she never dreamed it would be one of the major fiction works in the Year of the Cop.

"In 'Law and Order' I wanted to show policemen at home. Things that happen to you



Frank Serpico

also happen to them. It's the best book I've done (it's her fifth) because in this one they encouraged me to be a writer. I was told, 'Go ahead — write the fat book,' and I did."

She is constantly gesturing with her hands and she provides an unending commentary on cop life, fictional and real. But when the topic turns to police corruption, Dorothy hesitates.

"Sure, it's in the police department and part of the public's indignation stems from the fact policemen are upholders of the law. I'm not justifying corruption. I'm explaining it. What do you do? Make a collar or look the other way? You could wind up like Serpico . . . he got a bullet in the head."

What of Police Commissioner Patrick Murphy of New York?

"He's a damn good commissioner. He brought an aura of reasonableness to the department. I thought he set a terrific example in the shootout in Brooklyn. He exerted an enormous discipline there and the men went along with him."

Since "Law and Order" is about an Irish family of cops, what of the Irish Mafia in the department?

"It was just a tag. It doesn't exist. I worked with a lot of Irish cops and I'm part Irish. My father was a Goldstein and my mother was an O'Brien (her grandmother was an O'Malley) but I've got to admit the Irish are a close corporation. The class of 1940 changed that. You had Jews, Italians and, of course, the Irish, but 85 percent of the Class of 1940 in the academy was college educated. It was the most educated group and a problem at the same time. Some of the instructors hadn't even finished high school. One cop I knew had two Ph.D.s and he stayed on the job 24 years. He's a teacher now."

Women's Wear Daily

Dinosaur

Continued from page 19.

bones with parts that had been broken or eaten away by time.

Such bones are replaced or repaired in plaster or fiberglass and made to fit into their proper places in the skeleton.

"There are different ways of fabricating," Dr. Langston said. "The major procedure we used was to put two of the originals in position and then make a clay model of the one that would appear as the structural transition. In the fabricated piece we split the difference between the characteristics of the originals."

Dr. Pulley said the completed exhibit

at the museum will have the animal on a slope, as if he were walking out of a swampy pond, thus the incline of the platform—which will also be landscaped in simulation of that setting.

"A lot of work has been done on the pose and general appearance," Dr. Langston said. "I think we now have a pose that's satisfactory and, I guess, natural. We have to watch and not have an animal doing something that he could not have done in life."

"We paleontologists can run into a lot of trouble trying to infer what an animal could do. You couldn't tell from its skeleton, for example, that a goat can climb a tree."

Sunday, May 27, 1973. THE HOUSTON POST

BALLET RUDY JENKINS

6427 Atwell 667-3416
Classes All Summer — All Levels

Apply now for admission to
SUMMER CLASSES
MAY 28 THRU AUG. 17
AT
**HOUSTON BAPTIST
COLLEGE**

Contact the ADMISSIONS OFFICE 7502 Fondren Road,
Houston, Texas 77036 or call 774-7661 for information
and applications.

If you missed
the Science Fair,
don't miss

Winners
Weekend



47 exhibitors --the top award winners of the 1973 Science Engineering Fair of Houston --gather to meet you, to discuss and demonstrate their projects. Join us for a weekend of science and youth . . . and an education into the remarkable, industrious things these students are doing.

Museum of Natural Science
Hermann Brown Memorial Auditorium
Hermann Park

June 9 & 10 12 noon to 5 p.m. daily

The Science Engineering Fair of Houston is sponsored by
The Houston Post and the Engineers Council of Houston